



Fachbereich Rechts- und
Wirtschaftswissenschaften
Fachgebiet Entrepreneurship

Prof. Dr. Carolin Bock

Hochschulstraße 1
64289 Darmstadt

Tel. +49 6151 16 - 20925
Fax +49 6151 16 - 20014
bock@ent.tu-darmstadt.de

Certificate of performance in the module „Project in Entrepreneurship and Innovation Management“

Dear Ladies and Gentlemen,

Datum
10.12.25

During the summer term 2025, Mr. Raphael Northoff served as a dedicated project member in our course „Project in Entrepreneurship and Innovation Management“.

The project was focused on applying entrepreneurial methods in a real-world setting, accompanied by workshops on customer needs analysis, design thinking, corporate foresight, and business model development.

Mr. Northoff's group was tasked with conducting a market analysis and finding industry contacts for our associated partner institution, the Institute of Microwave Engineering and Photonics (IMP) of Technical University of Darmstadt and their invention, the Terahertz Photonic Vector Network Analyzer (TPVNA). Given the product's estimated 10-year time-to-market horizon, the group focused on early market validation.

The team's strategy to approach the task involved creating an online survey and personal interview outline, in collaboration with IMP, to gather firsthand market insights from engineers, researchers, and industry experts. To facilitate outreach and provide an accessible thematic introduction, Mr. Northoff proactively authored a one-page project summary about the TPVNA. This document became the essential foundation for contacting potential survey responders via LinkedIn and email.

Following the evaluation, the survey results enabled the group to determine a realistic product price, evaluate specific product features, and make recommendations for further development regarding customer needs. In



addition, the team conducted three insightful interviews and gained positive feedback from survey responders showing interest in collaborating with IMP.

Ultimately, the group's findings and recommendations were summarized in a detailed report and presented graphically for a final presentation delivered to the project supervisors and IMP.

Throughout the project, Mr. Northoff demonstrated precise analytical thinking and effective leadership capabilities. From an early phase, he took ownership of coordinating regular team meetings and motivated his teammates to successfully complete the project. His performance in the module was extraordinary, very motivated, and revealed highly professional skills. I wish him all the best for his further academic and personal journey.

Yours sincerely,

Prof. Dr. Carolin Bock